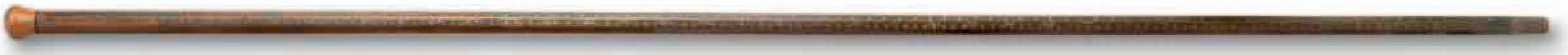
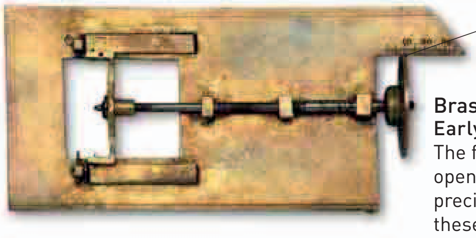
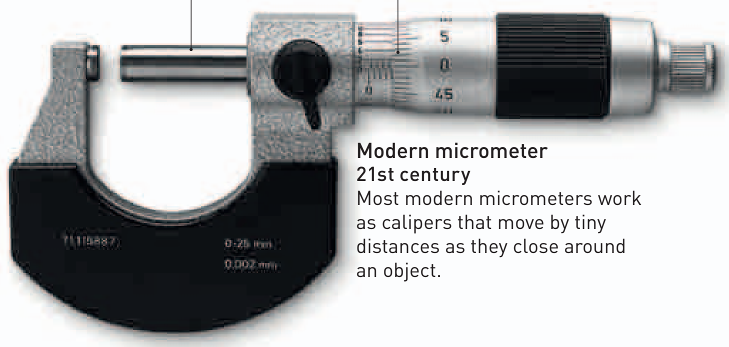


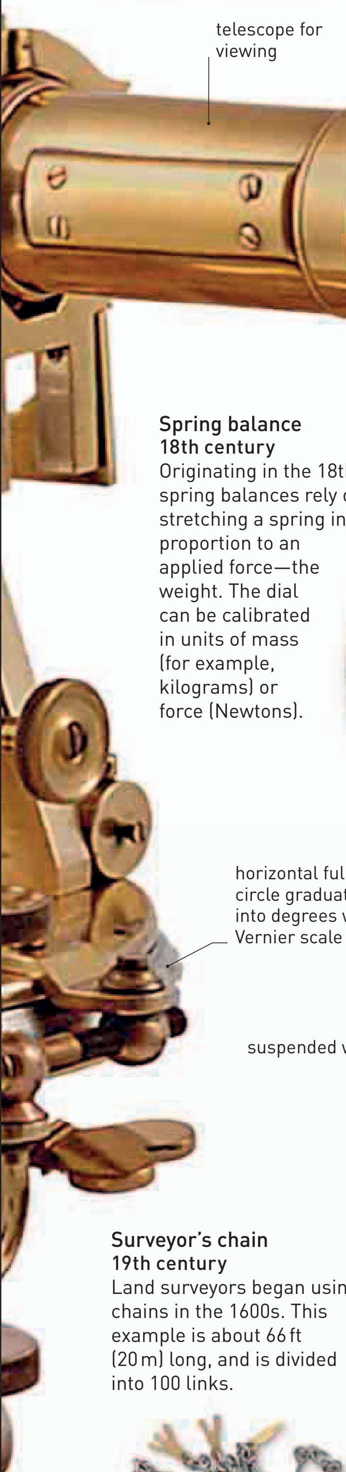


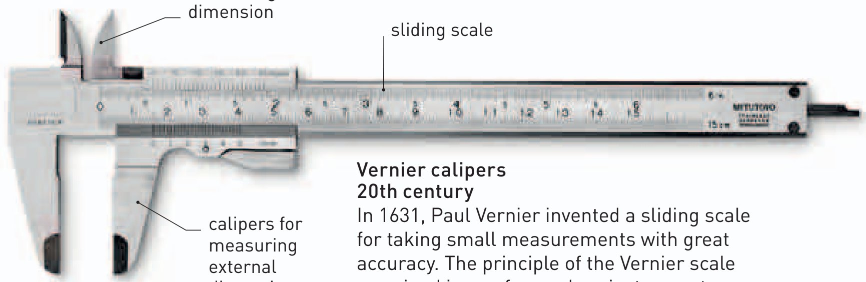
Brass micrometer
Early 19th century
The first micrometers opened up the field of precision engineering; these adjustable screwlike devices enable accurate measurement of small distances.



Modern micrometer
21st century
Most modern micrometers work as calipers that move by tiny distances as they close around an object.

Yardstick
18th century
The Imperial yard (3 ft or 0.9 m) has long been a popular unit of measurement for construction work. Most yardsticks can be used as rulers.





Vernier calipers
20th century
In 1631, Paul Vernier invented a sliding scale for taking small measurements with great accuracy. The principle of the Vernier scale remained in use for modern instruments.

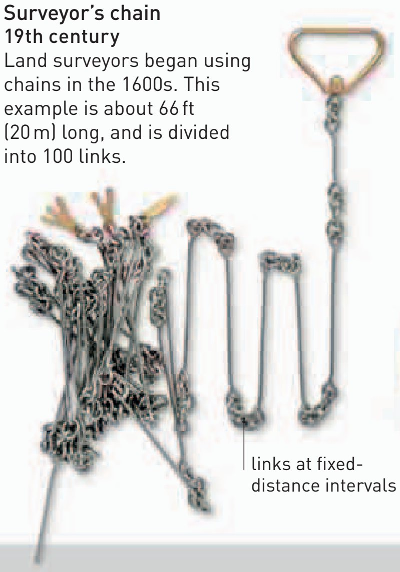
Laser distance meter
21st century
This shoots a laser pulse at a distant object and measures the time taken for the pulse to be reflected back.

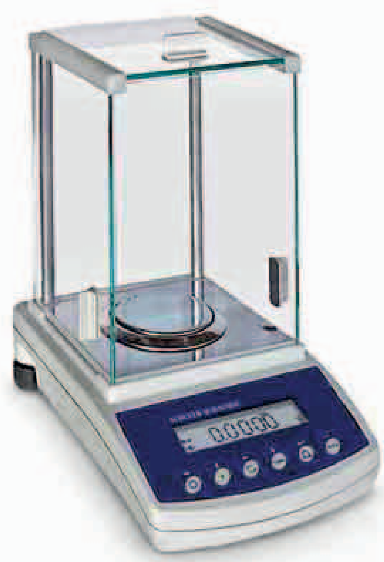
Spring balance
18th century
Originating in the 18th century, spring balances rely on stretching a spring in proportion to an applied force—the weight. The dial can be calibrated in units of mass (for example, kilograms) or force (Newtons).

Cased balance
18th century
Beam-balances were used in science and medicine, but small portable balances were also used for such purposes as measuring coins.

Weighing scales
18th century
Scales determine an unknown weight by counterbalancing it with known weights until equilibrium is achieved.

Surveyor's chain
19th century
Land surveyors began using chains in the 1600s. This example is about 66 ft (20 m) long, and is divided into 100 links.





Analytical balance
21st century
The most sophisticated modern digital balances can weigh minute fractions of a gram, and are so sensitive that they must be protected from vibrations, dust, and air movements.